

# PROJECT FOR THE INTRODUCTION TO ARTIFICIAL INTELLIGENCE (2026 SPRING)

The project process includes the following steps:

- A. Determination of project groups and choosing the project topics
- B. Assignment of topics to project groups
- C. Submission of projects and presentations

## A- DETERMINATION OF PROJECT GROUPS AND CHOOSING THE TOPICS

1. Students should communicate among themselves and form project groups consisting of at least 4 and at most 6 members. Each group will choose a group spokesperson to manage communications and upload project documents.
2. The project topics are listed at the end of the document.
3. After the group members and a spokesperson are determined, each group must rank all the available project topics (search problems) according to their preference priority.
4. To inform me about their group members and the prioritized topic list (all topics should be ordered according to the requested priority), the spokesperson of each group should send an e-mail to [baslan@yildiz.edu.tr](mailto:baslan@yildiz.edu.tr) in the format below by **Friday, April 24, 2026, at 22.00**. Emails must be sent before the specified deadline.
5. Groups may propose a different application topic that is not listed, provided that they obtain my approval.

\*\*\*\*\* A Sample Email \*\*\*\*\*

The subject of the E-mail: IAI\_2026\_PROJECT

### GROUP MEMBERS:

|                 |                                 |
|-----------------|---------------------------------|
| Name Surname 1, | Student Number 1 (spokesperson) |
| Name Surname 2, | Student Number 2                |
| Name Surname 3, | Student Number 3                |
| Name Surname 4, | Student Number 4                |
| Name Surname 5, | Student Number 5                |
| Name Surname 6, | Student Number 6                |

### THE SELECTED TOPICS THAT WERE ORDERED ACCORDING TO REQUESTED PRIORITY:

The topic 10  
The topic 8  
The topic 7  
The topic 1  
The topic 4  
The topic 5  
The topic 6  
The topic 13  
The topic 12  
The topic 11  
The topic 9  
The topic 2

The topic 14  
The topic 3

\*\*\*\*\* The end of the Sample Email \*\*\*\*\*

## B- ASSIGNMENT OF TOPICS TO PROJECT GROUPS

1. Based on the emails received from the groups (and if necessary, by drawing lottery), topics will be assigned so that each topic falls into a maximum of two groups. Project topics will be announced on **25 April 2026**.
2. After this announcement, the students who are not included in any group may form a group and choose remaining topics by **Thursday, April 30, 2026, at 16.00**, by contacting me. After this deadline, the students who are not included in any group and choose any topics will not have their projects evaluated.
3. Requests for member changes must be submitted with valid justification no later than **May 4, 2026**. No changes will be allowed after this date.

## C-SUBMISSION OF PROJECTS AND PRESENTATIONS

1. The project, the details of which are given below, and the presentation document must be uploaded to the system (online.yildiz.edu.tr) by **Thursday, 21 May 2026, at 22:00**. Only spokesperson needs to upload the project; other group members do not need to do so.  
**Important:** The system will close on Thursday, May 21, 2026, at 22:00. Do not attempt last-minute uploads.
2. Project presentations will be made during the course time on **22 May, 5 June and 12 June (in the last 3 weeks)**. Each group member must participate, and presentations should not exceed 20-25 minutes in total.
3. Since the algorithmic complexity and heuristic design vary by topic, grading will incorporate a 'difficulty factor.' Engineering effort on more challenging problems will be fairly rewarded.

## DETAILS OF THE PROJECT

C source code given in the attachment includes various heuristic and blind search methods for the path problem from a starting point to a destination point on the Romanian map, which was handled in the course. The details of code will also be discussed in the lectures on 24.04.2026. However,

before that, please review the source code and observe output results by building and running the code.

Based on the assigned topic, you should develop a new C program by updating the necessary places (**related to the topic of the problem and generalized A\* search**) in the C source code. Information about the places that need to be updated is given on the table below. The names, number and types of the input and output parameters of the functions in the source code must remain unchanged. You should only update the function blocks where modifications are required.

| File Name                  | Update                                                                                                                                |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <i>GRAPH_SEARCH.c</i>      | -                                                                                                                                     |
| <i>HashTable.c</i>         | 1. Generate_HashTable_Key( )                                                                                                          |
| <i>Standart_Search.c</i>   | 1. First_InsertFrontier_Search_TREE ( )<br>2. Insert_Priority_Queue_GENERALIZED_A_Star ( )                                            |
| <i>SpecificToProblem.c</i> | All functions (New functions can be added if necessary.)                                                                              |
| <i>data_types.h</i>        | 1. enum ACTIONS,<br>2. typedef struct State<br>(New variables and data types can be added if necessary.)                              |
| <i>GRAPH_SEARCH.h</i>      | 1. PREDETERMINED_GOAL_STATE<br>2. ACTIONS_NUMBER<br>3. MAX_SEARCHED_NODE<br>(New variables and data types can be added if necessary.) |
| <i>HashTable.h</i>         | 1. HASH_TABLE_BASED_SIZE<br>2. HASH_TABLE_INCREASING_RATE<br>3. MAX_KEY_SIZE                                                          |

**The following information should be included in the project report:**

1. The definition of the problem with all details (states, initial state, actions, transition model, etc.)
2. The definition of the heuristic function or heuristic functions that will be used
3. As a result of performing several simulations (for different problem sizes and initial states), the results obtained for each algorithm (the solution path, path cost, number of nodes produced, number of nodes in memory, computation time, etc.)
4. Sufficient screen shots and results should be supplied in the report.

The developed **C source codes (not including the exe file)**, the prepared **report** and the **presentation file** should be compressed and uploaded to [online.yildiz.edu.tr](http://online.yildiz.edu.tr) before the deadline.

You can contact me with your questions about the project and C source code.

## PROJECT TOPICS

**1. N-puzzle game:** Solve the game for the version of 8-puzzle, 15-puzzle, and 24-puzzle. The goal is to reach a specific target configuration by sliding tiles into the empty spot.  
(<https://www.mathsisfun.com/games/arrange.html>)

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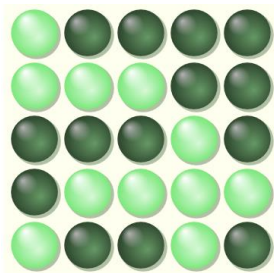
|   |   |   |
|---|---|---|
| 7 | 2 | 4 |
| 5 |   | 6 |
| 8 | 3 | 1 |

Start State

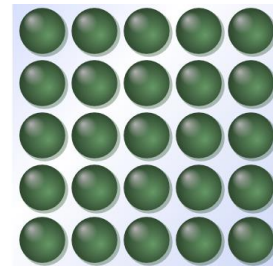
|   |   |   |
|---|---|---|
|   | 1 | 2 |
| 3 | 4 | 5 |
| 6 | 7 | 8 |

Goal State

**2. Turn All the Lights Out:** Some of the lights arranged on a matrix are on, while others are off. When you touch a light, if the light you touched and the horizontally and vertically neighboring lights are on, they turn off, and if they are off, they turn on. The game ends when all the lights are off. Solve the game for 4x4, 8x8 and 10x10 matrices.  
(<https://www.mathsisfun.com/games/allout.html>)

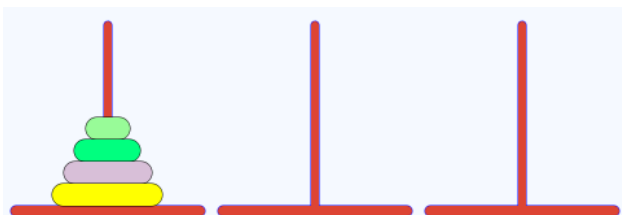


Start State



Goal State

**3. Hanoi Towers:** Solve the game for 3 disks, 5 disks, and 9 disks.  
(<https://www.mathsisfun.com/games/towerofhanoi.html>)

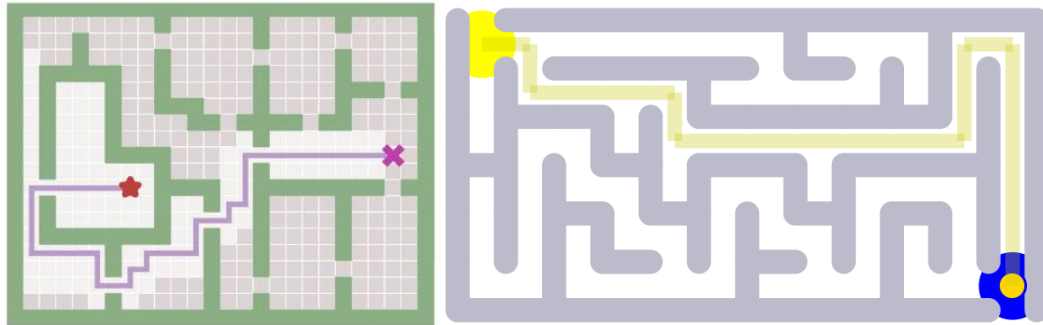


Start State

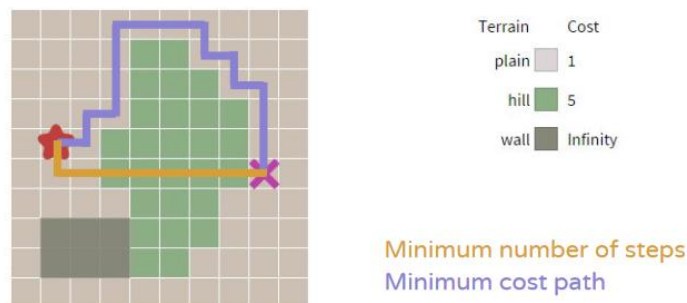


Goal State

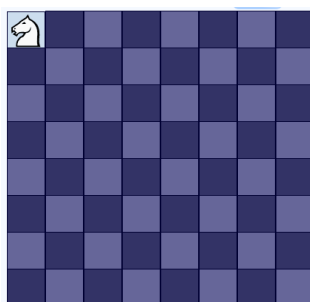
**4. The Shortest Path in Maze:** Finding the way from one point to another in a maze. Design two square (or rectangular) mazes, one medium-sized (at least 20x20) and one large (at least 50x50). Find solutions for random start and goal states in these mazes.  
(<https://www.mathsisfun.com/measure/mazes.html>)



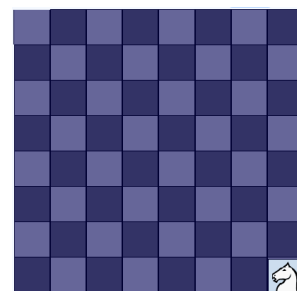
**5. The Shortest Path in Terrain:** Design two square (or rectangular) terrains, one medium-sized and one large. In these terrains, find solutions for random starting and goal states. Hilly areas, wetlands, and walls in the terrain should have different costs.



**6. Knight Game on the Chessboard:** In this game, the aim is to move a knight in a certain position on the chessboard to another intended point of the board with the minimum number of moves, by following the rules of the knight's movement. Solve the game for 8x8, 16x16 and 32x32 chessboards. Close some of the squares of the chessboard to prevent the knight from coming onto it. For example, let the knight cannot move on squares E3, A6, B2.



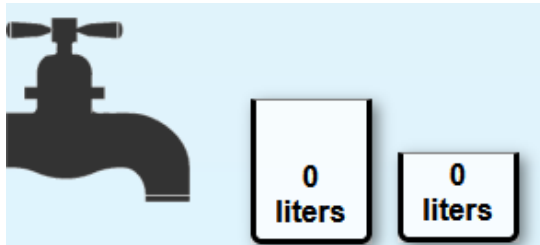
Start State



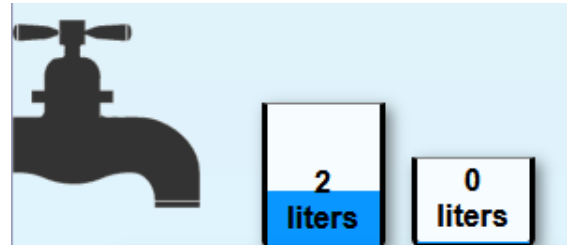
Goal State

7. **Jugs Puzzle Game:** There are 1 tap, 1 drain and 2 jugs in this game. When the jugs are filled from the tap, the entire jug is filled, and when the jug is emptied into the drain, the entire jug is emptied. When transferring water from one jug to another, the suitable amount of water is transferred, and the water is not wasted. The aim is to hold the exact amount of water in the larger jug. For example, in the example below, the aim is to hold exactly 2 liters of water in the larger jug when there are 5- and 3-liter jugs. Find solutions for two separate problems where there are 2 jugs and 3 jugs.

(<https://www.mathsisfun.com/games/jugs-puzzle.html>)



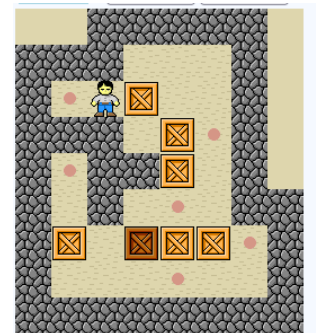
Start State



Goal State

8. **Sokoban Puzzle Game:** In a puzzle map, there is an agent (Sokoban) and one or more boxes. The purpose of the game is to push all the boxes to their predetermined goal positions. Here, the agent can only push the boxes; not pull them. The movement is restricted to four directions (up, down, left, and right), and the agent cannot pass through walls. Design two separate puzzle maps: one with a single box and another with two boxes.

(<https://www.mathsisfun.com/games/sokoban.html>)



9. **Word Ladder Puzzle:** The agent is given a start word and an end word. To win the game, the agent must change the start word into the end word progressively, creating an existing word at each step. Each step consists of a single letter substitution. The words should be in an English dictionary. Solve the puzzles using 3-letter words and 6-letter words.

([https://en.wikipedia.org/wiki/Word\\_ladder](https://en.wikipedia.org/wiki/Word_ladder))

10. **Rush Hour Problem (Traffic Jam Puzzle):** On a 6x6 grid filled with vehicles (cars and trucks), the agent must move interfering vehicles out of the way to allow a specific "red car" to reach the exit on the right. Vehicles can only move forward or backward along their orientation. The number of cars and trucks should be determined by the user.

([https://en.wikipedia.org/wiki/Rush\\_Hour\\_\(puzzle\)](https://en.wikipedia.org/wiki/Rush_Hour_(puzzle)))



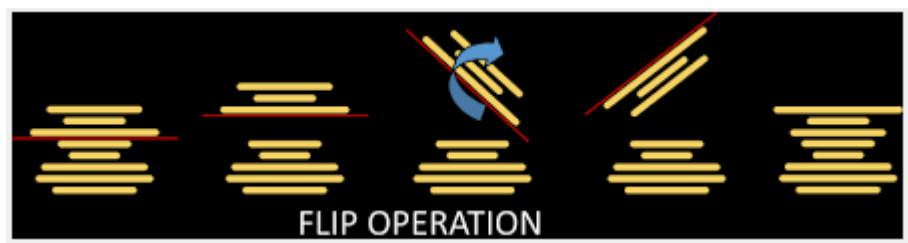
- 11. Generalized Missionaries and Cannibals (N-M, N-C, K-Capacity):** N Missionaries and N Cannibals must cross a river using a boat that can carry at most K people. At no point on either bank can the number of cannibals exceed the number of missionaries. Solve the problem for different N and K values.



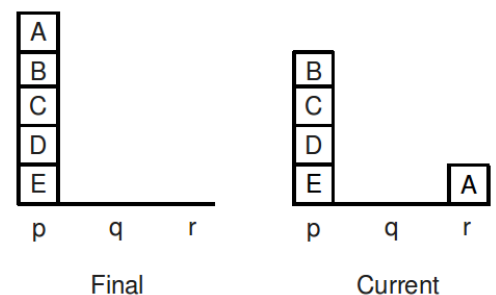
([https://en.wikipedia.org/wiki/Missionaries\\_and\\_cannibals\\_problem](https://en.wikipedia.org/wiki/Missionaries_and_cannibals_problem))

- 12. Pancake Sorting:** A stack of N pancakes of different sizes is given. The only allowed operation is a "flip": inserting a spatula at any point in the stack and reversing the order of all pancakes above it. The goal is to sort them by size. Solve the problem for different N values.

([https://en.wikipedia.org/wiki/Pancake\\_sorting](https://en.wikipedia.org/wiki/Pancake_sorting))



- 13. Blocks World:** Solve the game for 5 blocks, 7 blocks, and 9 blocks. A robotic arm operates on a table with a set of labeled blocks (e.g., A, B, C, D, E). The arm can pick up exactly one "clear" block (a block with nothing resting on top of it) and place it either directly on the table or on top of another "clear" block. The goal is to transform an initial random configuration into a specific target configuration (e.g., a single vertical stack spelling A-B-C-D-E).



([https://en.wikipedia.org/wiki/Blocks\\_world](https://en.wikipedia.org/wiki/Blocks_world))

- 14. Peg Solitaire:** A board contains holes and pegs in a specific pattern (usually a cross or plus shape). A valid move consists of a peg jumping orthogonally over an adjacent peg into an empty hole, which removes the jumped-over peg from the board. The goal is to reach a state where only one peg remains. ([https://en.wikipedia.org/wiki/Peg\\_solitaire](https://en.wikipedia.org/wiki/Peg_solitaire))

